Walker AI – Repair & Patch Logic Specification (Developer Reference)

This document defines how repair and patch work should be automatically estimated inside Walker AI. Repair work is typically required after demolition or when plumbing, electrical, or structural work requires wall or floor access. These rules allow the system to add realistic repair allowances without adding extra user questions.

Base Repair Pricing Table

Repair Item	Unit	Low	Average	High
Small drywall patch	EA	$80	$120	$180
Medium drywall patch	EA	$150	$250	$400
Large drywall repair	SQF	$6	$9	$14
Ceiling drywall patch	EA	$120	$200	$350
Subfloor patch repair	EA	$120	$200	$350
Plywood subfloor replacement	SQF	$6	$9	$14
Concrete floor patch	EA	$200	$350	$700
Tile patch repair	EA	$120	$200	$350
Tile replacement area	SQF	$12	$18	$25

Automatic Repair Triggers

The system should automatically add repair allowances when certain trades or demolition actions occur.

Trigger Condition	Recommended Allowance
Plumbing work affecting walls	$200 repair allowance
Electrical work requiring wall access	$150 repair allowance
Drain relocation	$350 floor repair allowance
Tile demolition	$200 tile patch allowance
Structural framing work	$500 repair allowance
Ceiling plumbing access	$200 ceiling patch allowance

Developer Logic Example

Example logic rule: if plumbing relocation occurs, automatically add drywall patch and floor repair allowances to the estimate.

Condition	Action
plumbing_relocation = true	add drywall_patch allowance
drain_relocation = true	add floor_patch allowance
electrical_scope = true	add drywall_patch allowance
tile_demolition = true	add tile_patch allowance

Implementation Goal

These rules allow Walker AI to automatically account for common renovation repairs without requiring additional questions from the user. This improves estimate accuracy and reduces change orders.